Task 1 Final Report
Customer Churn Prediction using Machine Learning

1. Objective

Develop a supervised machine learning model capable of predicting whether a customer is likely to churn. The project demonstrates the end-to-end ML pipeline including preprocessing, model training, evaluation, comparison, and model persistence.

2. Problem Statement

Customer churn is a major challenge for subscription-based businesses. Accurately identifying customers who are likely to leave enables organizations to improve retention through targeted interventions.

3. Dataset

The project uses the Telco Customer Churn dataset from Kaggle. It contains customer demographic information, account details, subscribed services, billing information, and the churn label (Yes/No).

4. Data Preprocessing

• Removed the customerID column.
• Converted TotalCharges into numeric values.
• Handled missing values using median imputation.
• Encoded categorical features using One-Hot Encoding.
• Standardized numerical features.
• Split the dataset into 80% training and 20% testing data.

5. Machine Learning Models

Two supervised classification algorithms were implemented:
• Logistic Regression
• Random Forest Classifier

Both models were trained using the same preprocessing pipeline and compared using identical evaluation metrics.

6. Model Evaluation

Performance was evaluated using:
• Accuracy
• Precision
• Recall
• F1-Score
• ROC-AUC Score
• 5-Fold Cross Validation

• Confusion Matrix
• ROC Curve

7. Results and Discussion

Both models achieved good predictive performance. Logistic Regression provided a strong baseline and was easy to interpret. Random Forest captured nonlinear relationships more effectively and generally produced better Accuracy, F1-Score, and ROC-AUC. Feature importance analysis indicated that tenure, contract type, monthly charges, and total charges were among the most influential predictors.

8. Tools and Technologies

Python, Google Colab, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Joblib.

9. Learning Outcomes

• Applied an end-to-end supervised machine learning workflow.
• Performed data preprocessing and feature engineering.
• Compared multiple classification algorithms.
• Interpreted evaluation metrics.
• Visualized model performance.
• Saved the trained model for future deployment.

10. Conclusion

The project successfully fulfilled all internship requirements by implementing a complete customer churn prediction system. After comparing both algorithms, Random Forest was selected as the preferred model due to its superior overall performance. This project strengthened practical skills in data preprocessing, supervised learning, model evaluation, visualization, and machine learning best practices.

11. Internship Requirements Checklist

Requirement	Status
Data Preprocessing	Completed
Train/Test Split	Completed
Cross Validation	Completed
Logistic Regression	Completed
Random Forest	Completed
Accuracy, Precision, Recall, F1, ROC-AUC	Completed
Confusion Matrix & ROC Curve	Completed
Model Comparison	Completed
Model Saving	Completed